HMS — Skills Interview Reference

Easy-English 2-line definitions + where each technology is used in the Hospital Management System (Laravel 12 + Livewire 3). Ready for recruiter follow-up.

Livewire 3

Definition: A Laravel library that makes pages interactive without writing JavaScript — the server re-renders only the changed part. You bind form fields and run actions with plain PHP methods.

Used in this project: Every admin panel (Patients, Rooms & Beds, Pharmacy, Billing, Settings, Features, Testimonials, tenant onboarding) is a Livewire component — real-time wire:model inputs, toggles and CRUD with zero page reloads.

Alpine.js

Definition: A tiny JavaScript library for small interactions — menus, popups, counters — written straight in the HTML. Livewire 3 bundles Alpine inside it.

Used in this project: Layouts call window.Alpine.initTree() after Livewire boots so every input keeps syncing; page behaviour is verified by an automated browser audit with 0 console errors.

REST API

Definition: A standard way to build web APIs where every URL is a resource and the HTTP verbs (GET / POST / PUT / PATCH / DELETE) say what to do. Data travels as JSON.

Used in this project: Built API v1 (routes/api/*) with a {success, message, data} envelope for the mobile app — appointments, patients, vitals, lab, pharmacy, billing and platform endpoints for the operator.

Laravel Sanctum

Definition: Laravel's built-in token authentication for APIs and single-page apps. Login swaps email + password + device name for a Bearer token.

Used in this project: POST /api/v1/auth/login issues one token per device; auth:sanctum guards every private endpoint. It also gates which roles may log in to the app (clinic = all roles, hospital = moving roles only).

RBAC (Role-Based Access Control)

Definition: The system checks a person\u2019s role before letting them act — a nurse cannot see billing, an accountant cannot order medicine. Access is granted by role + permission, not per user.

Used in this project: 10 roles with hms_can / hms_role_enabled checks and 403 aborts on every route and action; clinic mode automatically hides hospital-only roles and modules.

Multi-tenancy

Definition: One application serving many separate facilities, each with its own data, inside one database. A middleware works out which facility you belong to from the website address.

Used in this project: Shared schema with a tenant_id column plus global scopes; the ResolveTenant middleware maps host ? Tenant::current(); each facility gets its own subdomain or custom domain while the super-admin panel sits on a separate host.

Bootstrap 4 + AdminLTE

Definition: Ready-made HTML/CSS/JS building blocks — grids, cards, tables, menus — for a clean responsive admin interface. AdminLTE is a Bootstrap-based admin theme.

Used in this project: Power the admin layout, sidebar and the compact guest website; custom hms- classes built on top keep the small, consistent look across all pages.

jQuery

Definition: A small JavaScript library that makes DOM changes, events and AJAX shorter and simpler. Older but still the standard in admin themes.

Used in this project: Ships with the AdminLTE / Mix asset pipeline and handles the theme\u2019s UI helpers (widgets, toggles) alongside Livewire.

Laravel Mix

Definition: A build tool (Webpack underneath) that compiles SCSS and JS into browser-ready files. You run npm run dev or npm run prod.

Used in this project: package.json scripts compile the theme assets; keeps the compact-theme SCSS DRY and deployable.

DOMpdf

Definition: A PHP library that turns HTML + CSS into PDF files. Perfect for printable documents with no external service.

Used in this project: The Barryvdh/laravel-dompdf package powers the print-out centre: GST invoice, medicine slip (thermal size), case paper (A5) and prescription (A5) PDF downloads.

Intervention Image

Definition: A PHP image library to open, crop, resize and save pictures. Ideal for cleaning up uploads before they hit the server.

Used in this project: Resizes patient photos, doctor/department images, feature tiles and testimonial avatars on upload (e.g. 140×140 avatars) so the UI stays tidy and pages stay fast.

Jitsi Meet (WebRTC)

Definition: A free, open-source video-conferencing tool you can embed in a page. WebRTC delivers real-time video/audio in the browser.

Used in this project: Used for in-system video meetings: the Meeting model builds a stable meet.jit.si/room, the host starts the call and recording, and invited staff see a live countdown in the header.

Laravel Queues + Notifications

Definition: Queues run slow jobs in the background so the page never waits; Notifications send activity to users through one unified API — in-app, SMS, WhatsApp or email.

Used in this project: In-app notification bell for meeting invites, leave status, doctor alerts and appointment reminders; a scheduled command fires SMS/WhatsApp reminders best-effort before a patient slot.

PHPUnit

Definition: The standard testing framework for PHP. You write test cases and assert the expected result so features never silently break.

Used in this project: 270 tests / 1100+ assertions across feature tests (login, RBAC, billing, lab, mobile app access…), run automatically in CI on every push.

Git + GitHub

Definition: Version control: Git records every change as a commit; GitHub hosts the repository and shows the full history, branches and pull requests.

Used in this project: 55+ commits over the build lifecycle with clear feature-by-feature messages — a recruiter can read exactly how the system was built.

GitHub Actions (CI/CD)

Definition: GitHub's built-in automation: on every push a workflow installs dependencies, lints the code and runs the test suite so errors are caught automatically.

Used in this project: .github/workflows/ci.yml runs PHP lint + the full PHPUnit suite on every push — the project ships green after each change.

Responsive Web Design

Definition: Designing pages that look right on phones, tablets and desktops — columns, fonts and spacing adapt to the screen width with no horizontal scroll.

Used in this project: The compact theme is fully fluid; a browser audit (34 checks, 0 horizontal overflow) verifies public + admin pages, so the same panel works on a doctor\u2019s phone during rounds.